Airline Reservation System(Voltaire)

Package: default

Login

```
-MAX_ATTEMPTS: int
-MIN_PASSWORD_LENGTH: int
-NAVY: Color
-BLUE: Color
-airlineSystem: AirlineReservationSystem
-usernameField: JTextField
-passwordField: JPasswordField
-shortcutsOn: boolean

+Login(airlineSystem: AirlineReservationSystem)
-buildWelcomePanel(): JPanel
-buildLoginPanel(): JPanel
-columnPanel(background: Color, border: EmptyBorder): JPanel
-label(text: String, size: int, style: int, color: Color): JLabel
-button(text: String, size: int, style: int, foreground: Color, background: Color): JButton
-styleInput(field: JTextField): void
-passwordInput(): JPasswordField
-showMessage(parent: Component, title: String, message: String, type: int): void
-loginUser(): void
-endLoginSession(): void
-authenticate(username: String, password: String): User
-openDashboard(user: User): void
-wrongPassword(user: User): AirlineException
-createPassengerAccount(): void
-validateRegistration(username: String, email: String, password: String, confirm: String): void
-changePassword(user: User): void
-validateNewPassword(user: User, current: String, newPassword: String, confirm: String): void
-installShortcuts(): void
-addShortcuts(container: Container, used: ArrayList<Character>): void
-freeLetter(text: String, used: ArrayList<Character>): int
+main(args: String[]): void
```

Settings_ad

```
-role: String
-parent: JFrame
-airlineSystem: AirlineReservationSystem
-navy: Color
-blue: Color
-light: Color

+Settings_ad(parent: JFrame, role: String, airlineSystem: AirlineReservationSystem)
-createSettingsUI(): void
-createButton(text: String): JButton
```

AvailableEmails

```
-EMAILS: String[][]

+verify(email: String, password: String): boolean
```

Package: admin

AirlineReservationSystem

```
+HOME_COUNTRY: String
+ON_TIME: String
-FLIGHT_NO: int
-ROUTE: int
-DEPARTURE: int
-RETURN: int
-TIME: int
-PRICE: int
-CATEGORY: int
-STATUS: int
-REFUND: int
-LOCATION_COUNTRY: int
-LOCATION_STATUS: int
-users: ArrayList<User>
-bookingRecords: ArrayList<BookingRecord>
-flightCatalog: ArrayList<FlightClasses[]>
-notices: ArrayList<Notice>
-cancelledFlights: ArrayList<CancelledFlight>
-seatClasses: SeatClass[]
-passengerRegistry: PassengerRegistry
-userTableModel: DefaultTableModel
-flightTableModel: DefaultTableModel
-locationTableModel: DefaultTableModel
-bookingTableModel: DefaultTableModel
-currentUser: User
-logoutAction: ActionListener
-changePasswordAction: ActionListener
-settingsAction: ActionListener
-requestCounter: int
-syncingUsers: boolean

+AirlineReservationSystem()
-initTables(): void
-addFlightRow(flightNo: String, route: String, departure: String, returnDate: String, time: String,
  price: String, refundPolicy: String): void
-userRow(user: User): Object[]
+getUserTableModel(): DefaultTableModel
+getFlightTableModel(): DefaultTableModel
+getLocationTableModel(): DefaultTableModel
+getBookingTableModel(): DefaultTableModel
+getSeatClasses(): SeatClass[]
+getFlightCatalog(): ArrayList<FlightClasses[]>
+updateCatalogFromRow(row: int): void
-catalogIndexOf(flightNo: String): int
-createFlightClass(seatClass: SeatClass, row: int, origin: String,
  destination: String): FlightClasses
+removeFlight(flightNo: String): void
+getAvailableLocations(): String[]
+getOriginRoutes(): String[]
-locationLabel(row: int): String
-countryOf(place: String): String
+getFlightCategory(origin: String, destination: String): String
+isLocationUsed(place: String): boolean
+getLocationLabel(row: int): String
+getCancelReason(flightNo: String): String
+hasActiveBookings(flightNo: String): boolean
+cancelFlight(flightNo: String, reason: String): void
+cancelFlight(flightNo: String, reason: String, cancelledBy: String): void
+getUsers(): ArrayList<User>
+findUser(text: String): User
+generateUserID(): String
-userIdInUse(userId: String): boolean
+addUser(user: User): void
+registerPassengerAccount(userId: String, username: String, email: String,
  password: String): void
+deleteAccount(user: User): boolean
+getCurrentUser(): User
+setCurrentUser(currentUser: User): void
-installUserTableSync(): void
-syncUserFromRow(row: int): void
+usernameError(row: int, username: String): String
-cellText(model: DefaultTableModel, row: int, col: int): String
+getBookingRecords(): ArrayList<BookingRecord>
+addBookingRecord(record: BookingRecord): void
+findBookingRecord(referenceId: String): BookingRecord
+getBookingRecordsOf(userId: String): ArrayList<BookingRecord>
+getTotalRevenue(): double
+countActiveBookings(userId: String): int
+submitCancellationRequest(record: BookingRecord): void
+markBookingCancelled(referenceId: String): void
+updatePaymentStatus(referenceId: String, paymentStatus: String): void
+refreshBookingTable(): void
-bookingRow(record: BookingRecord): Object[]
+generateReferenceID(): String
+generateBookingID(): String
-uniqueCode(prefix: String, column: int): String
-codeInUse(code: String, column: int): boolean
+createPassengerId(): String
+addNotification(userId: String, message: String): void
+takeNotifications(userId: String): ArrayList<String>
+notifyAdmins(message: String): void
+showNotifications(): void
+setLogoutAction(logoutAction: ActionListener): void
+setChangePasswordAction(action: ActionListener): void
+getChangePasswordAction(): ActionListener
+setSettingsAction(settingsAction: ActionListener): void
+getSettingsAction(): ActionListener
+logout(): void
```

FlightClasses

```
-seatClass: SeatClass
-flightNumber: String
-origin: String
-destination: String
-departureDate: String
-returnDate: String
-departureTime: String
-baseFare: double
-refundPolicy: String
-category: String
-availableSeats: int

+FlightClasses(seatClass: SeatClass, flightNumber: String, origin: String, destination: String,
  departureDate: String, returnDate: String, departureTime: String, availableSeats: int,
  baseFare: double, refundPolicy: String, category: String)
+getClassName(): String
+calculateFare(): double
+getCheckedBaggage(): String
+getMeal(): String
+getFlightNumber(): String
+getOrigin(): String
+getDestination(): String
+getDepartureDate(): String
+getReturnDate(): String
+getDepartureTime(): String
+getRefundPolicy(): String
+getAvailableSeats(): int
+isDomestic(): boolean
+returnsOn(date: String): boolean
```

SeatClass

```
-name: String
-fareMultiplier: double
-baggage: String
-meal: String

+SeatClass(name: String, fareMultiplier: double, baggage: String, meal: String)
+defaults(): SeatClass[]
+getName(): String
+getFareMultiplier(): double
+getBaggage(): String
+getMeal(): String
+setPerks(baggage: String, meal: String): void
```

Notice

```
-userId: String
-message: String

+Notice(userId: String, message: String)
+getUserId(): String
+getMessage(): String
```

AirlineException

```
-title: String
-endsSession: boolean

+AirlineException(message: String)
+AirlineException(title: String, message: String, endsSession: boolean)
+getTitle(): String
+endsSession(): boolean
```

InputTools

```
-InputTools()
+allDigits(text: String): boolean
+capitalLetters(text: String, length: int): boolean
+isValidEmail(email: String): boolean
+isExpiryDate(text: String): boolean
+money(amount: double): String
+requiredLabel(text: String): JLabel
+twoDecimals(amount: double): String
+zeroPad(number: int, digits: int): String
+joinText(items: ArrayList<String>): String
```

AdminDashboardPanel

```
-app: AirlineReservationSystem
-userTableModel: DefaultTableModel
-flightTableModel: DefaultTableModel
-locationTableModel: DefaultTableModel
-bookingTableModel: DefaultTableModel
-FONT_LABEL: Font
-FONT_BUTTON: Font
-FONT_TABLE: Font
-FONT_TABLE_HEADER: Font

+AdminDashboardPanel(app: AirlineReservationSystem)
-blueButton(text: String): JButton
-BubSort(model: DefaultTableModel, priceColumnIndex: int, ascending: boolean): void
-restoreDefaultOrder(model: DefaultTableModel): void
-LinearSearch(table: JTable, model: DefaultTableModel, searchColIndex: int, labelName: String,
  keyword: String): void
-performUniversalEdit(table: JTable, model: DefaultTableModel): void
-performFlightEdit(table: JTable, model: DefaultTableModel): void
-askFlight(current: String[], ignoredRow: int): String[]
-routePart(route: String, index: int): String
-placeBox(places: String[], selected: String): JComboBox<String>
-flightError(values: String[], ignoredRow: int): String
-isPositiveNumber(text: String): boolean
-formatPrice(text: String): String
-flightNoAlreadyExists(flightNo: String, ignoredRow: int): boolean
-styleTable(table: JTable): JScrollPane
-wrapTab(table: JTable, bottomActionPanel: JPanel, model: DefaultTableModel,
  searchColIndex: int, searchLabel: String, isFlightTab: boolean): JPanel
-wrapTabWithTopRightActions(table: JTable, bottomActionPanel: JPanel,
  model: DefaultTableModel, searchColIndex: int, searchLabel: String, isFlightTab: boolean,
  customEdit: ActionListener): JPanel
-verifyAdminPassword(enteredPassword: String): void
-cancelFlightWithBookings(flightNo: String): void
-askAdminPassword(action: String): boolean
-createManageUsersTab(): JPanel
-createManageFlightsTab(): JPanel
-createSeatClassesTab(): JPanel
-seatClassRow(seatClass: SeatClass): Object[]
-editPerks(seatClass: SeatClass): boolean
-createManageLocationsTab(): JPanel
-addLocation(): void
-locationCodeExists(code: String, ignoredRow: int): boolean
-editLocation(table: JTable): void
-createManageBookingsTab(): JPanel
-performBookingEdit(table: JTable): void
-createReportsTab(): JPanel
-createCard(title: String, value: String): JPanel
```

BookingRecord

```
-referenceId: String
-bookingId: String
-userId: String
-flight: FlightClasses
-seats: ArrayList<String>
-passengerNames: ArrayList<String>
-passengerIds: ArrayList<String>
-paymentMethod: String
-paymentAccount: String
-paymentStatus: String
-amountPaid: double
-bookingStatus: String
-cancellationStatus: String
-refundStatus: String
-requestId: String
-passengerDetails: ArrayList<PassengerDetails>

+BookingRecord(referenceId: String, bookingId: String, userId: String, flight: FlightClasses,
  seats: ArrayList<String>, passengerNames: ArrayList<String>,
  passengerIds: ArrayList<String>, paymentMethod: String, paymentAccount: String,
  paymentStatus: String, amountPaid: double)
+setPassengerDetails(passengers: PassengerDetails[]): void
+getPassengerEmailsText(): String
+getPassportNumsText(): String
+getPassengerIdsText(): String
+setPaymentStatus(paymentStatus: String): void
+getReferenceId(): String
+getBookingId(): String
+getUserId(): String
+getFlight(): FlightClasses
+getSeats(): ArrayList<String>
+getPaymentMethod(): String
+getPaymentAccount(): String
+getPaymentStatus(): String
+getAmountPaid(): double
+getBookingStatus(): String
+setBookingStatus(bookingStatus: String): void
+getCancellationStatus(): String
+setCancellationStatus(cancellationStatus: String): void
+getRefundStatus(): String
+setRefundStatus(refundStatus: String): void
+getRequestId(): String
+setRequestId(requestId: String): void
+getFlightNumber(): String
+getOrigin(): String
+getDestination(): String
+getTicketType(): String
+isPaidOnline(): boolean
+isActive(): boolean
+getPassengerNamesText(): String
+getSeatsText(): String
```

User

```
+PASSENGER: String
+STAFF: String
+ADMIN: String
+ROLES: String[]
-userId: String
-username: String
-fullName: String
-email: String
-password: String
-role: String
-status: String
-loginAttempts: int

+User(userId: String, username: String, fullName: String, email: String, password: String,
  role: String, status: String)
+User(userId: String, fullName: String, email: String, password: String, role: String,
  status: String)
+usernameOf(email: String): String
+matchesLogin(text: String): boolean
+isRole(roleName: String): boolean
+isBlocked(): boolean
+canBeBlocked(): boolean
+getUserId(): String
+getUsername(): String
+setUsername(username: String): void
+getEmail(): String
+setEmail(email: String): void
+getPassword(): String
+setPassword(password: String): void
+getRoleName(): String
+setRole(role: String): void
+getStatus(): String
+setStatus(status: String): void
+getLoginAttempts(): int
+setLoginAttempts(loginAttempts: int): void
```

PassengerRegistry

```
-counter: int

+createNewId(): String
```

CancelledFlight

```
-flightNumber: String
-reason: String

+CancelledFlight(flightNumber: String, reason: String)
+getFlightNumber(): String
+getReason(): String
```

FlightDates

```
+NO_RETURN: String

+hasReturn(returnDate: String): boolean
+checkDeparture(departure: String): String
+checkReturn(departure: String, returnDate: String, required: boolean): String
+check(departure: String, returnDate: String, returnRequired: boolean): String
```

Package: passenger

Main_1

```
-headerPanel: JPanel
-messagePanel: JPanel
-bookingPanel: JPanel
-mainPanel: JPanel
-settings: JButton
-searchButton: JButton
-subtitle: JLabel
-logo: JLabel
-title: JLabel
-tripTypeBox: JComboBox<String>
-originBox: JComboBox<String>
-destinationBox: JComboBox<String>
-passengerBox: JComboBox<String>
-departureDateField: JTextField
-returnDateField: JTextField
-airlineSystem: AirlineReservationSystem
-selectedTripType: String
-selectedReturnDate: String
-selectedOrigin: String
-selectedDestination: String
-selectedDepartureDate: String
-selectedPassengerCount: int

+Main_1(system: AirlineReservationSystem)
+showNotifications(): void
+showBookingPage(): void
-deleteMyAccount(): void
+getAirlineSystem(): AirlineReservationSystem
+getSelectedTripType(): String
+getSelectedReturnDate(): String
-createFieldPanel(labelText: String, field: JComponent): JPanel
+resetBookingFields(): void
+main(args: String[]): void
```

PassengerDeets

```
-passengers: PassengerDetails[]
-passengerCount: int
-currentPassenger: int
-main: Main_1
-previousPage: SelectSeat
-chosenFlight: FlightClasses
-selectedSeats: ArrayList<String>
-givenNametext: JTextField
-surnameT: JTextField
-genderBox: JComboBox<String>
-birthT: JTextField
-emailT: JTextField
-contactT: JTextField
-passportT: JTextField

+PassengerDeets(main: Main_1, previousPage: SelectSeat, chosenFlight: FlightClasses,
  passengerCount: int, selectedSeats: ArrayList<String>)
+getChosenFlight(): FlightClasses
+getSelectedSeats(): ArrayList<String>
+getPassengers(): PassengerDetails[]
```

BookingConfirmation

```
+BookingConfirmation(main: Main_1, chosenFlight: FlightClasses,
  selectedSeats: ArrayList<String>, passengers: PassengerDetails[], totalPrice: double,
  paymentMethod: String, paymentStatus: String, referenceNumber: String, bookingID: String)
-addDetail(panel: JPanel, label: String, value: String): void
-loadBackgroundImage(): Image
```

«interface»
Payable

```
+processPay(amount: double): boolean
+hasEnoughBalance(amount: double): boolean
```

PaymentMethods

```
+getPayment(): String
+processPay(amount: double): boolean
+hasEnoughBalance(amount: double): boolean
+getAccountNumber(): String
+refundToAccount(accountNumber: String, amount: double): boolean
+getBalance(): double
+getBalanceOf(accountNumber: String): double
```

eWallet

```
+verifyAccount(mobileNumber: String, accountName: String, password: String): boolean
```

GCashPayment

```
-accounts: PaymentAccount[]

+getPayment(): String
#getAccounts(): PaymentAccount[]
```

OTCPayment

```
+getPayment(): String
+processPay(amount: double): boolean
+hasEnoughBalance(amount: double): boolean
```

PaymentRefund

```
-methodNamed(paymentMethod: String): PaymentMethods
+refund(paymentMethod: String, accountNumber: String, amount: double): boolean
+balanceOf(paymentMethod: String, accountNumber: String): double
```

PassengerDetails

```
-givenName: String
-surname: String
-birth: String
-passportNum: String
-email: String
-contactNum: String
-gender: String
-passengerId: String

+getGivenName(): String
+getSurname(): String
+getPassportNum(): String
+getEmail(): String
+getContactNum(): String
+getPassengerId(): String
+setPassengerId(passengerId: String): void
+setGivenName(givenName: String): void
+setSurname(surname: String): void
+setBirth(birth: String): void
+setPassportNum(passportNum: String): void
+setEmail(email: String): void
+setContactNum(contactNum: String): void
+setGender(gender: String): void
```

Settings

```
-navy: Color
-blue: Color
-parent: JFrame
-role: String
-changePasswordAction: ActionListener
-bookingsAction: ActionListener
-logoutAction: ActionListener
-deleteAccountAction: ActionListener

+Settings(parent: JFrame, role: String, changePasswordAction: ActionListener,
  bookingsAction: ActionListener, logoutAction: ActionListener)
+Settings(parent: JFrame, role: String, changePasswordAction: ActionListener,
  bookingsAction: ActionListener, logoutAction: ActionListener,
  deleteAccountAction: ActionListener)
-createSettingsPage(): void
-createButton(text: String): JButton
```

FlightSelection

```
-selectedOrigin: String
-selectedDest: String
-selectedDate: String
-chosenFlight: FlightClasses
-main: Main_1
-passengerCount: int
-airlineSystem: AirlineReservationSystem
-selectedTripType: String
-selectedReturnDate: String
-selectedClass: String
-chosenReturnFlight: FlightClasses

+FlightSelection(main: Main_1, origin: String, destination: String, date: String,
  passengerCount: int)
-matchesOutbound(flight: FlightClasses): boolean
-findReturnFlight(flights: ArrayList<FlightClasses[]>, outbound: FlightClasses): FlightClasses
-showMessage(panel: JPanel, message: String): void
+getChosenFlight(): FlightClasses
+getSelectedTripType(): String
+getSelectedReturnDate(): String
```

SelectSeat

```
-chosenFlight: FlightClasses
-totalSeats: int
-seatBtns: JButton[]
-selectedSeats: ArrayList<JButton>
-availSeats: seatManager
-continueBtn: JButton
-main: Main_1
-previousPage: FlightSelection
-passengerCount: int
-seatConfirmed: boolean

+SelectSeat(main: Main_1, chosenFlight: FlightClasses, previousPage: FlightSelection,
  passengerCount: int)
+createSeatBtns(): JPanel
+getSelectedSeats(): ArrayList<String>
+confirmSeat(): boolean
+releaseSelectedSeats(): void
-NavigationButtons(): JPanel
```

PaymentPage

```
-main: Main_1
-totalPrice: double
-selectedPayment: PaymentMethods
-df: DecimalFormat
-detailsLayout: CardLayout
-paymentDetails: JPanel
-previousPage: PassengerDeets
-chosenFlight: FlightClasses
-selectedSeats: ArrayList<String>
-passengers: PassengerDetails[]
-bookingCompleted: boolean

+PaymentPage(main: Main_1, chosenFlight: FlightClasses, selectedSeats: ArrayList<String>,
  passengers: PassengerDetails[], previousPage: PassengerDeets)
-payWithWallet(mobileNumber: String, accountName: String, password: String): void
-confirmTax(): boolean
-showWarning(message: String, title: String): void
-processElectronicPayment(): void
-finalizeBooking(paymentStatus: String): void
-showConfirmation(paymentMethod: String, paymentStatus: String, referenceNumber: String,
  chosenFlight: FlightClasses, selectedSeats: ArrayList<String>,
  passengers: PassengerDetails[], bookingID: String): void
-recordAdminBooking(referenceNumber: String, paymentStatus: String): String
```

«interface»
Refundable

```
+refundToAccount(accountNumber: String, amount: double): boolean
+getBalanceOf(accountNumber: String): double
```

OnlinePayment

```
#verifiedAccount: PaymentAccount

#getAccounts(): PaymentAccount[]
#verifyLogin(number: String, accountName: String, password: String): boolean
-findAccount(accountNumber: String): PaymentAccount
+getAccountNumber(): String
+getBalance(): double
+getBalanceOf(accountNumber: String): double
+hasEnoughBalance(amount: double): boolean
+processPay(amount: double): boolean
+refundToAccount(accountNumber: String, amount: double): boolean
```

CardPayment

```
-accounts: PaymentAccount[]

+getPayment(): String
#getAccounts(): PaymentAccount[]
+verifyAccount(cardNumber: String, password: String): boolean
```

MayaPayment

```
-accounts: PaymentAccount[]

+getPayment(): String
#getAccounts(): PaymentAccount[]
```

PaymentAccount

```
-accountNumber: String
-accountName: String
-password: String
-balance: double

+PaymentAccount(accountNumber: String, accountName: String, password: String,
  balance: double)
+checkAccount(accountNumber: String, password: String): boolean
+checkAccount(accountNumber: String, accountName: String, password: String): boolean
+getAccountNumber(): String
+getBalance(): double
+hasEnoughBalance(amount: double): boolean
+withdraw(amount: double): boolean
+deposit(amount: double): void
```

seatManager

```
-bookedSeats: ArrayList<String>

-seats(flight: FlightClasses, seatNumber: String): String
+isSeatBooked(flight: FlightClasses, seatNumber: String): boolean
+bookSeat(flight: FlightClasses, seatNumber: String): boolean
+releaseSeat(flight: FlightClasses, seatNumber: String): boolean
+releaseSeats(flight: FlightClasses, seatNumbers: ArrayList<String>): boolean
```

MyBookings

```
-bookingTable: JTable
-model: DefaultTableModel
-airlineSystem: AirlineReservationSystem
-records: ArrayList<BookingRecord>

+MyBookings(airlineSystem: AirlineReservationSystem)
-loadBookings(): void
-requestCancellation(): void
```

Package: staff

StaffDashboard

```
-navy: Color
-blue: Color
-light: Color
-bookingTable: JTable
-flightTable: JTable
-cancellationTable: JTable
-paymentTable: JTable
-bookingModel: BookingTableModel
-flightModel: NonEditableTableModel
-cancellationModel: NonEditableTableModel
-paymentModel: PaymentTableModel
-searchField: JTextField
-tabs: JTabbedPane
-paymentSortField: JComboBox<String>
-paymentSortOrder: JComboBox<String>
-bookingReview: BookingReview
-flightCancellation: FlightCancellation
-updatingBookingStatus: boolean
-updatingPaymentStatus: boolean
-paymentReferenceCounter: int
-flightCapacity: int
-settingsAction: ActionListener
-airlineSystem: AirlineReservationSystem

+StaffDashboard()
+StaffDashboard(airlineSystem: AirlineReservationSystem)
-loadSharedData(): void
-countBookedSeats(flightNo: String): int
-hasFlight(flightNo: String): boolean
+setSettingsAction(settingsAction: ActionListener): void
-createDashboard(): void
-createBookingsPanel(): JPanel
-linearSearch(keyword: String, startRow: int): int
-searchBookings(): void
-viewPassengerDetails(): void
-createFlightsPanel(): JPanel
-createCancellationPanel(): JPanel
-createPaymentsPanel(): JPanel
-generatePaymentReference(): String
-sortPayments(): void
-isOutOfOrder(before: String, key: String, ascending: boolean): boolean
-makePanel(): JPanel
-createButton(text: String, color: Color): JButton
-styleTable(table: JTable): void
+main(args: String[]): void
```

BookingTableModel

```
+BookingTableModel(columns: String[])
+isCellEditable(row: int, column: int): boolean
```

NonEditableTableModel

```
+NonEditableTableModel(columns: String[])
+isCellEditable(row: int, column: int): boolean
```

BookingStatusCellEditor

```
+BookingStatusCellEditor(choices: JComboBox<String>)
```

BookingReview

```
-parent: JFrame
-bookingModel: DefaultTableModel
-cancellationModel: DefaultTableModel
-flightModel: DefaultTableModel
-system: AirlineReservationSystem
-processor: CancellationProcessor

+BookingReview(parent: JFrame, bookingModel: DefaultTableModel,
  cancellationModel: DefaultTableModel, flightModel: DefaultTableModel,
  system: AirlineReservationSystem)
+reviewRequest(selectedViewRow: int, approved: boolean, cancellationTable: JTable): void
+confirmCancellationByBooking(bookingRef: String): boolean
-approveCancellation(bookingRow: int, requestRow: int, bookingRef: String, flightNo: String,
  eligibility: String): void
-isFlightCancelled(flightNo: String): boolean
-returnSeats(flightNo: String, count: int): void
-findBookingRow(bookingRef: String): int
```

FlightCancellation

```
-parent: JFrame
-flightModel: DefaultTableModel
-bookingModel: DefaultTableModel
-cancellationModel: DefaultTableModel
-system: AirlineReservationSystem
-processor: CancellationProcessor

+FlightCancellation(parent: JFrame, flightModel: DefaultTableModel,
  bookingModel: DefaultTableModel, cancellationModel: DefaultTableModel,
  system: AirlineReservationSystem)
-askStaffPassword(): boolean
+cancelEntireFlight(selectedViewRow: int, flightTable: JTable): void
-findRequest(bookingRef: String): int
-generateRequestID(): String
```

CancellationProcessor

```
-system: AirlineReservationSystem
-paymentRefund: PaymentRefund
-seats: seatManager
-money: DecimalFormat

+CancellationProcessor(system: AirlineReservationSystem)
+findRecord(system: AirlineReservationSystem, bookingRef: String): BookingRecord
+approveRequest(record: BookingRecord): String
+cancelByAdmin(record: BookingRecord): String
+rejectRequest(record: BookingRecord): void
+cancelByStaff(record: BookingRecord, reason: String): String
-cancel(record: BookingRecord, refundAllowed: boolean): String
-adminRefundNote(refundStatus: String): String
-refundMessage(record: BookingRecord, refundStatus: String): String
```

PaymentTableModel

```
+PaymentTableModel(columns: String[])
+isCellEditable(row: int, column: int): boolean
```

StatusCellRenderer

```
+getTableCellRendererComponent(table: JTable, value: Object, isSelected: boolean,
  hasFocus: boolean, row: int, column: int): Component
```